

Spectroscopic Signature of r -process Nucleosynthesis in a Gamma-Ray Burst: Resonant Deuterium Absorption in GRB 210704A

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Neutron star mergers are confirmed catastrophic stellar endpoint events such as binary neutron star mergers and collapsars are proposed to be sites of rapid neutron-capture process (r -process) nucleosynthesis, typically inferred from late-time kilonova emissions. Here, we r -process nucleosynthesis. We report the detection of a statistically significant spectral gap between sub-GeV and GeV photons in the prompt emission of GRB 210704A. By constraining the bulk Lorentz factor ($\Gamma \approx 200 - 300$), we map this gap to an energy of $\sim 2 - 7$ MeV around $3.5 - 6.7$ MeV in the comoving frame. We rule out alternative explanations and show that this feature is naturally explained by the resonant photodissociation of deuterium (^2H), whereas the Giant Dipole Resonance of heavier nuclei occurs at much higher energies (~ 20 MeV). Furthermore, the extremely short dynamical timescale of the prompt emission favors the survival of deuterium, as the formation of heavier r -process r -process elements is bottlenecked by slower β -decays. This observation provides the first spectroscopic evidence of light r -process r -process element formation within a relativistic jet, linking the immediate aftermath of a compact object merger the relativistic outflow generated in catastrophic stellar endpoints to nucleosynthesis.

Introduction.—The merger of—Core-collapse supernovae

(CC SNe) and binary neutron stars (BNS) or a neutron star and a black hole (NSBH) creates an extremely mergers both create neutron-rich environment, ideal for the rapid neutron-capture process (r -process) (e.g., see Refs. [1–6]). While environments; they are the two leading candidates for producing most of the rapid neutron capture process (r -process) elements in the universe [e.g. 1–10]. Although the gravitational wave event GW170817 and its associated kilonova AT2017gfo confirmed that such mergers synthesize heavy elements [4, 11, 12][4, 11], observational evidence has largely been restricted to the thermal emission of radioactive decay on timescales of days. Investigating nucleosynthesis during the prompt phase (minutes after the merger) remains a significant challenge, primarily due to intense radiation masking nuclear spectral lines.

Gamma-ray bursts (GRBs) launch relativistic jets that may entrain neutron-rich material from the central engine; a high-entropy high-enthalpy winds and fireballs associated with GRBs have been proposed to be a source of light-element synthesis [13–17][13–18]. If nuclear species survive in the jet, they imprint absorption features on the gamma-ray spectrum. However, identifying these features requires precise knowledge of the bulk Lorentz factor (Γ) to translate the observed energies to the comoving frame. In this Letter, we present the discovery of a discrete spectral gap in the high-energy emission of GRB 210704A. We demonstrate that this anomaly corresponds to the photodissociation threshold of deuterium

(^2H) in the jet’s comoving frame. This identification implies that the jet was loaded with r -process seed nuclei synthesized on second timescales, consistent with a neutron star merger origin.

Observation and Spectral Anomaly.—GRB 210704A was detected by the Fermi Gamma-ray Burst Monitor (GBM) and the Large Area Telescope (LAT) [19, 20]. Arguments from previous work on the unusual stellar progenitor of GRB 210704A [19, 21] are presented. The recent detailed multiwavelength analysis of GRB 210704A determines that the most likely redshift is $z = 2.34$ [22] by confirming Ly α absorption from line-stacking analysis with afterglow spectrum. In the following analysis, we use $z = 2.34$ to estimate the Lorentz factor and to determine the energetics of the comoving frame.

The burst exhibits a distinct separation in its high-energy photon distribution. While the MeV emission is bright, the GeV emission (> 100 MeV) arrives with a slight delay almost simultaneously and, crucially, displays a “zone of avoidance” in the energy spectrum. (Color online) (a) Left Y-axis with the unit of counts rate (s^{-1}): the histograms are the light curves of GBM detectors (NaI 5 and BGO 0, upper plot) and LAT (bottom plot); the high-energy photons are from the LAT data of TRANSIENT020 class (or evelass=16), and used in the analysis to maximize the statistics for spectral analysis of GRB prompt emission. Right Y-axis with the unit of MeV in bottom plot: energies of high-energy photons (denoted by circles). The gray shadow in the first 2 s denotes intersection of all gaps between between sub-GeV (blue circles) and GeV (red circles) photons. (b) The time-integrated spectral modeling with PL function for high-energy photons, with and without considering zero count in the gap (marked in gray).

As shown in FIG. 1(a), the photons are densely clustered in the sub-GeV regime (< 0.62 GeV) and the multi-GeV regime

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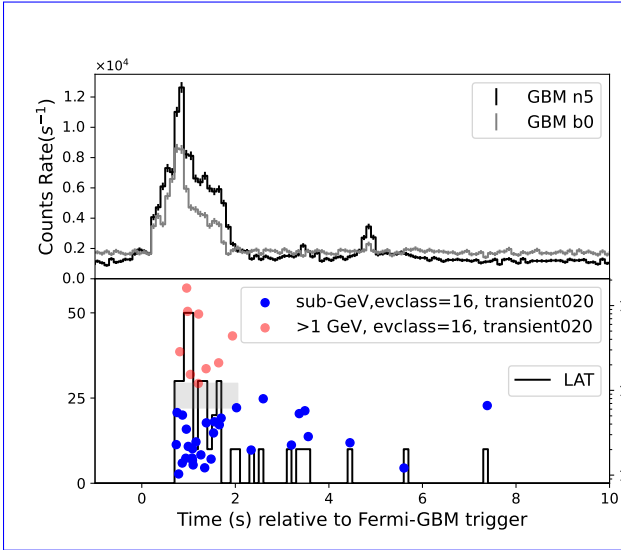


FIG. 1. (Color online) Left Y-axis with the unit of counts rate (s^{-1}); the histograms are the light curves of GBM detectors (NaI 5 and BGO 0, upper plot) and LAT (bottom plot); the high-energy photons are from the LAT data of TRANSIENT020 class (or evclass=16), and used in the analysis to maximize the statistics for spectral analysis of GRB prompt emission. Right Y-axis with the unit of MeV in bottom plot: energies of high-energy photons (denoted by circles). The gray shadow in the first 2 s denotes intersection of all gaps between sub-GeV (blue circles) and GeV (red circles) photons.

(> 1.21 GeV), leaving a statistically significant gap (denoted in gray shadow) between approximately 0.62 GeV and 1.21 GeV during the prompt phase (from 0.7 s to 2.0 s after trigger in GBM).

We quantified the

A full likelihood analysis is performed with GTLIKE, to quantify the statistical significance of this gap using a Monte Carlo (MC) simulation. Assuming. Similarly to [23], we compared a continuous power-law (PL) distribution that connects the sub-GeV and GeV components, we generated 1.4×10^5 synthetic photon samples folded with the detector response. The probability of observing a gap width ≥ 590 MeV (the intersection of all gaps in GeV emission in the first 2 seconds) in the time-integrated spectrum purely due to Poisson fluctuations was determined to be $p\text{-value} < 10^{-3}$; this $p\text{-value}$ is overestimated due to the smoothing effect of the time-integrated spectrum on the gap. Considering the range of the gap that changes over time, model with a model explicitly incorporating a spectral gap [24], and the analysis yields a preference for the gap model with $\Delta\text{BIC} = -2\Delta\log(\text{likelihood}) = 9.2$. This value constitutes strong statistical evidence against a simple continuous spectrum [25], further supporting the physical separation of the product of a sequence of $p\text{-values}$ ($\prod p\text{-value}$) for the corresponding larger gaps in time-resolved spectra is very tiny ($\ll 10^{-6}$, see details in Appendix ??) two emission components. Furthermore, a time-resolved analysis reveals that the upper energy envelope of the sub-GeV photons

and the lower energy envelope of GeV photons evolve synchronously, maintaining the gap structure. This strong correlation ($\rho > 0.7$, Spearman rank; see the details in Appendix A) suggests that the two components are physically linked but separated by an absorption mechanism.

To rigorously validate this spectral structure, we performed parameter estimation using Markov Chain Monte Carlo (MCMC) sampling with the EMCEE package [26]. Similarly to [23], we compared a continuous power-law model with a model explicitly incorporating a spectral gap using the Bayesian Information Criterion (BIC). As illustrated in FIG. 1 (b), the analysis yields a preference for the gap model with $\Delta\text{BIC} \approx 5.7$. This value constitutes positive statistical evidence against a simple continuous spectrum [27], further supporting the physical separation of the two emission components.

Comoving Frame Energetics.—To interpret the physical origin of the gap, we must determine the Lorentz factor Γ . The early prompt MeV emission (< 0.5 s) is well-described by a non-dissipative photospheric model, allowing us to constrain the jet properties. A combined model of multicolor black-body (mBB, e.g. see [28–30]) plus a BAND-PL function [31] (with a break energy at $\sim 10^4$ keV) for the broad energy band is used to describe the spectrum of prompt emission from keV to GeV energy band; with a lowest BIC and difference $\Delta\text{BIC} > 10 \Delta\text{BIC} = 8.1$, it is statistically preferred over the other models (see details in Figure 3 in Appendix B). Using the method described in [32, 33], we derived Γ from the quasi-thermal component of the spectrum. The estimated Lorentz factor evolves with time, peaking at $\Gamma \approx 300$.

With $z = 2.34$, the Lorentz Factor is estimated to be ≥ 500 around the peak of luminosity and decaying to $\Gamma \approx 190$ one second later flux with the photon of highest energy (16.09 GeV); the model-dependent method using the quasi-thermal component in the prompt phase proposes a range of 500 to 900 (see the details in Appendix C).

Applying the Doppler correction $E' \approx E_{\text{obs}}(1 + z)/2\Gamma$, where $z \approx 0.27$ is the redshift derived from the Amati relation and jet modeling [21] with $\Gamma = 300$, the observed gaps translate gap ($[0.62, 1.21]$ GeV) translates to a comoving energy range of:

$$E'_{\text{gap}} \approx 23.5 - 76.7 \text{ MeV}. \quad (1)$$

This energy range is critical. We first rule out standard electromagnetic mechanisms. A two-component jet model (a fast spine and slow sheath, see, e.g., Ref. [34]) fails to explain the synchronized temporal evolution of the sub-GeV and GeV components. Similarly, Klein-Nishina suppression in Inverse Compton scattering would produce a high-energy cutoff rather than a discrete gap followed by a recovery in flux at > 1.2 GeV. Therefore, we turn to photon-matter interactions.

Resonant Nuclide Absorption.—We propose that the gap arises from the resonant absorption of photons by nuclei entrained in the jet. The comoving energy of ~ 4 MeV aligns remarkably well with the photodissociation threshold of deu-

155 terium:

$$\gamma + {}^2\text{H} \rightarrow p + n. \quad (2)$$

156 The cross-section for photodissociation peaks typically
157 around 4 – 5 MeV [35], which perfectly matches our derived
158 comoving energy E'_{gap} .

159 Crucially, this energy scale distinguishes deuterium from
160 all other potential nuclear absorbers. Heavier nuclides, par-
161 ticularly the Fe-group elements often invoked in supernova
162 models, interact with photons primarily through the Giant
163 Dipole Resonance (GDR). The GDR peak energy for nuclei
164 with mass number A is approximately $E_{\text{GDR}} \approx 31.2A^{-1/3} +$
165 $20.6A^{-1/6}$ MeV [36], which places the resonance at 15 – 25
166 MeV for heavy elements (e.g., ${}^{56}\text{Fe}$) [37]. Such absorption
167 would manifest at observed energies > 3 GeV, far above
168 the gap detected in GRB 210704A. The unique low-energy
169 threshold of deuterium makes it the only viable candidate for
170 an absorption feature at ~ 4 MeV.

171 If the resonant deuterium absorption almost fix the Lorentz
172 factor being around 300, we choose $\Gamma = 10^{2.5}$ as a typical
173 value in the following.

174 Timescales and Nucleosynthesis. —The— With the
175 standard internal shock model for the prompt emission, the
176 emitting radius is determined to be [38]

$$R \simeq 2\Gamma^2 c \delta t \simeq 6 \times 10^{12} \Gamma_{2.5}^2 \delta t_{-3} / (1+z) \text{ cm}, \quad (3)$$

177 To extract the time scale (δt) of pulse duration, a wavelet
178 decomposition is performed with the data of the internal shock
179 emission (photons of energy < 20 keV are required, and the
180 contribution from quasi-thermal emission is 7%. The light
181 curve has a sampling rate of 4000 Hz; see the details of the
182 method in [39]). δt is determined to be about 1 ms with an
183 uncertainty of a few ms. Considering the redshift ($z = 2.34$),
184 the emitting radius $R \sim 2 \times 10^{12}$ cm.

185 The identification of deuterium is further supported by the
186 dynamical timescales of the GRB jet. The dynamical time in
187 the comoving frame is extremely short, on the order of hun-
188 dreds of seconds:

$$t'_{\text{dyn}} \approx \frac{2\Gamma t_{\text{obs}} / (1+z)}{\Gamma c} \sim \frac{3000.1 R_{12} \Gamma_{2.5}^{-1}}{\Gamma c} \text{ s}. \quad (4)$$

189 where $t_{\text{obs}} \sim 1$ s is the time in the observer's frame. In a
190 neutron-rich outflow ejected from a merger, nucleosynthesis
191 begins with the rapid capture of neutrons by protons ($p + n \rightarrow$
192 ${}^2\text{H}$). To form heavier r -process elements (up to Lanthanides),
193 the chain must proceed through the creation of seed nuclei.
194 However, the formation of seeds heavier than deuterium is
195 rate-limited by weak interactions (β -decays), which typically
196 require timescales of seconds to minutes to proceed signifi-
197 cantly. On the other hand, the r -process in the non-relativistic
198 outflow is not influenced by such a jet effect, and lanthanides
199 and other heavy elements still form [12]. 0.1 s is too short for
200 heavier elements to form.

201 Given the relativistic expansion, the jet material is “frozen
202 out” before significant quantities of heavy elements can form

203 via β -decay. Consequently, the jet composition at the pho-
204 tospheric radius is expected to be dominated by nucleons
205 and light isotopes, specifically deuterium, rather than heavy
206 metals. This kinetic constraint robustly ~~excludes Fe-group~~
207 ~~elements and~~ reinforces the deuterium interpretation.

208 *Optical depths for photon-electron and photon-deuteron*
209 *interactions in relativistic internal shocks.*— The comoving
210 density of the jet, mainly contributed by protons (with a rest
211 mass of m_p), is expressed in terms of the isotropic ~~isotropic~~
212 ~~kinetic energy (matter luminosity~~ ($L_K = E_K / T_{\text{dur,lab}}$; E_K
213 ~~is the kinetic energy, $T_{\text{dur,lab}}$ is the duration time in the lab~~
214 ~~frame) of the jet at the non-thermal emission radius R as [27]~~

$$n'_p = \frac{E_K}{4\pi R^3 m_p c^2} \frac{L_K}{4\pi R^2 c \Gamma^2 m_p c^2} = \frac{1.71.8}{\Gamma_{2.5}^3} \times 10^{17} \frac{E_{K,53}}{L_{K,55}} R_{12.5}^{-3} \Gamma_{2.5}^{-2}$$

215 where $E_{K,53} = E_K / 10^{53} \text{ erg}$, $L_{K,55} = L_K / 10^{55} \text{ erg s}^{-1}$ and
216 the unit of R is cm; in the following, the convention $Q_s =$
217 $Q / 10^s$ is adopted in cgs units throughout the text. ~~The With~~
218 ~~the~~ comoving density of deuterium n'_D ~~is estimated with and~~
219 the number fraction of deuterium (Y_D) ~~to be:~~

$$n'_D = 0.1 n'_p Y_{D,-1}.$$

220 ~~The, the~~ optical depth for the photodissociation of deuterium
221 is

$$\tau_{\gamma D} \sim \frac{\sigma_{\gamma D} n'_D R}{\Gamma} = \frac{1.1 E_{K,53} R_{12.5}^{-2} 1.4 Y_{D,-1} \Gamma_{2D} L_{K,55} R_{12}^{-1} \Gamma_{2.5}^{-3}}{\Gamma}, \quad (6)$$

222 where $\sigma_{\gamma D} \simeq 2 \times 10^{-27} \text{ cm}^2$ at a pho-
223 ton energy $\epsilon'_\gamma \simeq 4 \text{ MeV}$, $\epsilon'_\gamma \simeq 4.48 \text{ MeV}$ in the deuteron rest
224 frame [35]. ~~Note that $\Gamma \simeq 200$ is taken for the merged shells~~
225 ~~in internal shocks; it is typically less than the maximum value~~
226 ~~estimated by the quasi-thermal component.~~

227 On the other hand, the GeV photons should be freely
228 escaped from the scattering of electrons. Internal shocks are
229 natural sites for the dissipation of the kinetic energy of a bary-
230 onic fireball [40–43]. In the internal shock model of gamma-
231 ray bursts, electrons accelerated at ~~mildly~~ relativistic shocks
232 are commonly assumed to follow a non-thermal distribution
233 [44].

$$N(\gamma_e) \propto \gamma_e^{-p}, \quad \gamma_e \geq \gamma_m, \quad (7)$$

234 where γ_m is the minimal Lorentz factor of the electrons ac-
235 celerated in the internal shocks. For typical internal shock pa-
236 rameters adopted in the literature, $\gamma_m \sim 10^{3.5} - 10^6$ [45, 46].
237 The scattering regime between photons and relativistic elec-
238 trons is determined by

$$x \equiv \frac{\gamma_e \epsilon'_\gamma}{m_e c^2}, \quad (8)$$

239 where ϵ'_γ is the photon energy in the comoving frame. One has
240 $x \gtrsim 2.5 \times 10^4$ for $\epsilon'_\gamma = 4 \text{ MeV}$ and $\gamma_m \gtrsim 10^{3.5}$, $x \gtrsim 8 \times 10^5$
241 for $\epsilon'_\gamma \sim 4 \text{ MeV}$ and $\gamma_m \simeq 10^5$ as typical values. In this

Klein-Nishina limit, the total scattering cross section is suppressed to [47]

$$\sigma_{\text{KN}} \simeq \frac{3}{8} \sigma_T \frac{\ln(2x) + 1/2}{x} \lesssim 1.146 \times 10^{-28-30} \text{ cm}^2, \quad (9)$$

thus, the optical depth for Compton scattering in the Klein-Nishina regime is about

$$\tau_{\text{KN}} \sim \frac{\sigma_{\text{KN}} n'_e R}{\Gamma} \lesssim 0.6 E_{\text{K},53} 2.6 \times 10^{-3} L_{\text{K},55} R_{12.5}^{-2} \Gamma_2^{-1} = 0.5 Y_D, \quad (10)$$

with assuming local electrical neutrality in the jet, $n'_e \simeq n'_p$. The break energy (ϵ_b) in the spectrum of the non-thermal component described by a BAND function (e.g. the synchrotron emission by the relativistic electrons in magnetic field) may correspond to $\tau_{\text{KN}}(\epsilon'_b) \sim 1$ at $\gamma_e \sim \gamma_m$. From the results of the modeling of the non-thermal component, the break energy is $\epsilon_b \sim 10^4$ keV, which corresponds to $\gamma_m \sim 10^{5.4}$ in this case; $\sigma_{\text{KN}} \sim 10^{-30} \text{ cm}^2$ is much lower for the photons of $\epsilon'_\gamma = 4$ MeV, due to a deeper Klein-Nishina regime.

The estimations above show that the GeV photons at the gap should be absorbed by the deuterium while the other photons are able to escape from the scattering of the electrons.

Discussion.—Let us demonstrate the plausibility for of the observation of photodissociation of deuterium in the jet of GRB 210704A as follows: the key point is whether baryon loading in a relativistic outflow could produce enough deuterium; second, GeV photons are emitted while those in the resonant regions are absorbed by deuterium, which requires that $\tau_{\gamma D}$ and τ_{KN} (for GeV photons) should satisfy

$$\tau_{\text{KN}} < 1 \lesssim \tau_{\gamma D}. \quad (11)$$

The freezeout Y_D can be greater than a few percent or even as large as 10% under the right conditions below the radius of GeV emissions (see the simulation in Ref. [17]).

The total energy of the outflow could be estimated by the radiative $E_{\gamma, \text{iso}}$ and the radiative efficiency (ϵ_γ) as

$$E_{\text{tot}, \text{iso}} = E_{\gamma, \text{iso}} / \epsilon_\gamma; \quad (12)$$

for baryonic fireball, $E_{\text{K}} \lesssim (1 - \epsilon_\gamma) E_{\text{tot}, \text{iso}} L_{\text{K}} \lesssim (1 - \epsilon_\gamma) L_{\text{tot}, \text{iso}}$. It is reasonable that $\epsilon_\gamma \lesssim 10\%$, and this is because the radiation efficiency of internal shocks is typically low (e.g., see Refs. [38, 48]). $E_{\gamma, \text{iso}} \simeq 10^{52} \text{ erg}$ for GRB 210704A, $L_{\gamma, \text{iso}} = 1.0 \times 10^{54} \text{ erg s}^{-1}$ for the component from keV to GeV energy band (about 2.6 times that of the peaking component in MeV emission); therefore, we assume that E_{K} is comparable to 10^{53} erg . The internal shock radius is

$$R \simeq 2\Gamma^2 c \delta t \simeq 6 \times 10^{12} \Gamma_{2.5}^2 \delta t_{-3} \text{ cm},$$

which has an order of magnitude of $10^{12} - 10^{13} \text{ cm}$, because the time scale (δt) of pulse duration is determined to be about 1 ms by wavelet decomposition (with a sampling rate of 4000 Hz; see the details of the method in [39]); it lies in the typical

range for short GRBs [49]. Thus, with given $Y_D \simeq 0.1$, $E_{\text{K}} \sim 10^{53} \text{ erg}$ and $R \sim 10^{12.5} \text{ cm}$, equation $10^{55} \text{ erg s}^{-1}$. In fact, $L_{\gamma, \text{iso}}$ obtained from the modeling is time-averaged; the density of baryons is not uniform, and L_{K} for each pulse (or shell) is even larger.

Y_D can be as large as $\simeq 10\%$ under the right conditions in neutron-ion decoupling (see the simulation of Ref. [17]). The large production of deuterium could be through the spallation of α -particles [18], which is caused by neutron-ion decoupling, as well as internal shocks. In the intense internal shocks that occur in the jet of GRB 210704A, although nucleosynthesis is sensitive to these unknown details inside, we expect that Y_D might be even greater due to the spallation of α -particles. Considering that the fraction of α -particles could be comparable with those of nucleons for the short dynamic time [17, 18], $Y_D \simeq 0.3$ is not impossible and Eq (11) is satisfied. The baryon-loading mass in the jet is calculated to be $f_b M_{\text{iso}} \sim 10^{-6} M_\odot$ by $E_{\text{K}} = \Gamma M_{\text{iso}} c^2$ (the typical beaming factor for short GRBs is $f_b \sim 6 \times 10^{-3}$, see [49]), which is a reasonable value for merger events [4, 50, 51] in this case. For the Lorentz Factor, $\Gamma \simeq 300$ is a plausible value that could lead to $\tau_{\gamma D} \gtrsim 1$. Given $\gamma_m \simeq 5 \times 10^6$ and $p = 2$ of the power-law distribution of γ_e , for photons of much smaller energy, e.g. $E_{\text{obs}} \simeq 10$ keV, this satisfies $\tau_{\text{KN}} < 1$. This is well consistent with the observed prompt spectrum.

For the high-energy photons, because of the very small optical depth for Compton scattering, photons of PL component are emitted out of the outflow once produced. The interactions between high-energy photons and their target photons with lower energies are in the relativistic regime; thus the Lorentz factor of the outgoing pairs $\gamma_\pm \gg 1$ (where $\gamma_\pm \equiv \frac{E_0}{m_e c^2}$ and E_0 is the incoming photon energy in the center-of-momentum frame), the cross section for pair production is suppressed to be [52, 53],

$$\sigma_{\gamma\gamma} = \frac{3}{8} \sigma_T \gamma_\pm^{-2} (\ln 2\gamma_\pm^2 - 1) \simeq 10^{-31} \text{ cm}^2 \ll \sigma_T. \quad (13)$$

$\tau_{\gamma\gamma} \lesssim 1$ of the high-energy photons of 1.21 GeV–4.39 GeV which consist of the upper envelope of the gap, requires that $\Gamma \gtrsim 300$ at $R \sim 10^{12} \text{ cm}$, which is consistent well with our hypothesis.

Conclusion.—We identified a statistically significant spectral gap in the prompt emission of GRB 210704A. Interpreting this as resonant absorption by deuterium nuclei in a jet with $\Gamma \sim 200 - 300$, $\Gamma \simeq 300$ for the first time, we find a consistent physical picture that links the burst to a neutron-rich progenitor. This represents the first direct spectroscopic detection of nucleosynthesis ingredients in the r -process in the prompt gamma-ray phase, opening a new window for studying nuclear astrophysics in extreme environments.

Previous studies of GRB 210704A noted its peculiar properties and suggested a compact object merger origin, despite its duration lying in the intermediate range [19, 21]. Our finding provides a completely independent confirmation of this progenitor channel [19, 21, 22]. Unlike kilonovae, which probe the sub-relativistic ejecta days after the event, this

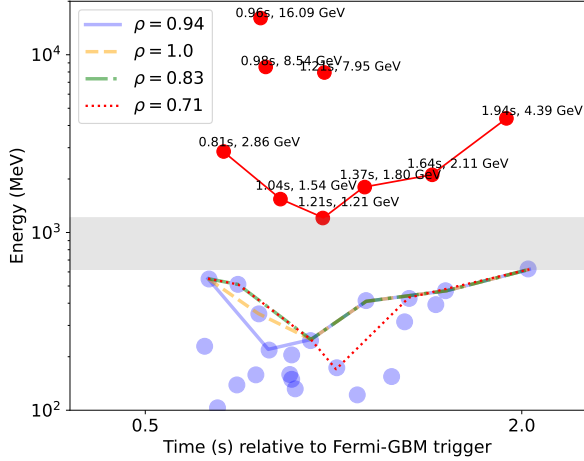


FIG. 2. The Spearman rank correlation coefficients of temporal evolution between $\gamma_{\text{GeV}}^{\text{envelope}}$ and $\gamma_{\text{sub-GeV}}^{\text{envelope}}$ (denoted by different line styles and colors) extracted with different bin-edge division ratios. The gray shadow between 0.62 and 1.21 GeV shows the intersection of all gaps between sub-GeV and GeV photons.

spectral feature probes the relativistic jet itself seconds after launch, revealing the initial chemical composition of the r -process flow. It should be noted that such observations are expected to be rare. GRB 210704A is a ‘short’ long burst that favors the abundance of deuterium; the high baryon loading of the outflow and rapid variability of the central engine create conditions for the occurrence of intense internal shocks.

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Appendix A: The possible correlation between $\gamma_{\text{sub-GeV}}^{\text{envelope}}$ and $\gamma_{\text{GeV}}^{\text{envelope}}$

The Spearman rank correlation coefficient (ρ) [54][27] is used to quantify the monotonic relationship of the temporal evolution between the energy of $\gamma_{\text{GeV}}^{\text{envelope}}$ (denotes the lower envelope of GeV photons) and that of $\gamma_{\text{sub-GeV}}^{\text{envelope}}$

(denotes the upper energy envelope of sub-GeV photons). $\gamma_{\text{GeV}}^{\text{envelope}}$ is naturally formed by the GeV photons as shown in FIG. 2 in red circles connected by a red line. In order to ensure a one-to-one correspondence between $\gamma_{\text{sub-GeV}}^{\text{envelope}}$ and $\gamma_{\text{GeV}}^{\text{envelope}}$, the maximum sub-GeV energy value in each time bin was extracted as $\gamma_{\text{sub-GeV}}^{\text{envelope}}$; the data binning in the range of $T_0 + [0.70, 2.05]$ s is performed as follows: 1) the edges of the bin are determined to be the n -equal division points in time of each adjacent GeV point in $\gamma_{\text{GeV}}^{\text{envelope}}$; 2) to reduce the impact of artificial selection, the bin-edge division ratio (rate) is tested from 0.01 to 0.99 with an increment of 0.01 (for example, a rate equal to 0.2 means left five-division point).

From all binning results with different bin-edge division ratios, four different energy envelopes of sub-GeV photons were extracted, as shown in FIG. 2. ρ is determined to be > 0.7 conservatively from results of all sets of bin edges, which implies a strong correlation between $\gamma_{\text{sub-GeV}}^{\text{envelope}}$ and $\gamma_{\text{GeV}}^{\text{envelope}}$ [55].

Appendix B: Spectral Modelings with different models and the analysis for the prompt emission

FIG. 3 shows the time-integrated spectral modeling results with mBB + BAND, BAND, blackbody (BB) mBB + PL and BAND + PL. The parameters of the three best models are listed below (BIC and χ^2 to the degree of freedom are shown in the figures).

- BAND+PL: for the BAND component, $\alpha = -0.06^{+0.14}_{-0.11}$, and $\beta = -2.80^{+0.26}_{-0.35}$, $E_p = 256.1^{+13.4}_{-16.1}$ keV, $F_{\text{BAND}} = 10.7^{+0.7}_{-0.7} \times 10^{-6}$ erg cm $^{-2}$ s $^{-1}$; for the PL component: index $p = -1.80^{+0.18}_{-0.09}$, $F_{\text{PL}} = 20.2^{+3.7}_{-4.2} \times 10^{-6}$ erg cm $^{-2}$ s $^{-1}$.
- mBB+PL: for the mBB component, $m = -0.29^{+0.37}_{-0.37}$, $kT_{\text{min}} = 19.1^{+3.0}_{-3.5}$ keV, $kT_{\text{max}} = 146.0^{+25.2}_{-17.3}$ keV, $F_{\text{mBB}} = 8.5^{+0.4}_{-0.3} \times 10^{-6}$ erg cm $^{-2}$ s $^{-1}$; for the PL component: $p = -1.81^{+0.06}_{-0.03}$, $F_{\text{PL}} = 22.4^{+5.6}_{-5.6} \times 10^{-6}$ erg cm $^{-2}$ s $^{-1}$.
- mBB+BAND: for the mBB component, $m = -0.37^{+0.55}_{-0.60}$, $kT_{\text{min}} = 18.8^{+3.7}_{-5.3}$ keV, $kT_{\text{max}} = 145.1^{+20.0}_{-14.1}$ keV, $F_{\text{mBB}} = 7.7^{+0.4}_{-0.4} \times 10^{-6}$ erg cm $^{-2}$ s $^{-1}$; for the BAND component: $\alpha = -1.68^{+0.04}_{-0.02}$, $\beta = -1.90^{+0.16}_{-0.07}$, $\log_{10}(E_p) = 3.89^{+0.17}_{-0.14}$ keV, $F_{\text{BAND}} = 23.3^{+1.7}_{-1.8} \times 10^{-6}$ erg cm $^{-2}$ s $^{-1}$.

From the residual distribution of 40 keV to 300 keV, there exists a systematic bias in the modeling with BAND + PL functions. The change on BIC is $\Delta\text{BIC} = 8.1$, which shows a strong preference for mBB + PL. Therefore, the mBB function is better than the BAND function for describing the

MeV peak component from the modeling results. In addition, the low-energy photon index for the peaking component is about -0.1, although it cannot be decisive evidence for the quasi-thermal emission. The ΔBIC of mBB+BAND to mBB+PL is 1.8, and BAND is not an unconventional model for the component of the broad energy band; thus, the mBB + BAND model is not better than mBB + PL.

It is evident that there exist two different components in the prompt emission. There are two candidate processes to interpret the observation:

1) There exist seed photons in the outflow, and some of them become the component of a broad energy band from keV to GeV via inverse Compton (IC) scattering by relativistic electrons. However, not all are emitted, especially for those with lower energies, which correspond to a larger scattering cross section with electrons; moreover, as electrons are cooled, the cross section of electron-photon scattering becomes larger and even increases to the Thomson cross section ($\sigma_T > \sigma_{KN}$). They will be thermalized below the photosphere and emitted when the radius reaches R_{ph} ; R_{ph} satisfies (r_0 is the initial radius for jet launching, and ranges from 10^7 cm to 10^9 cm) [53]

$$\begin{aligned} R_{ph} &\simeq \left(\frac{L_K \sigma_{e\gamma}}{8\pi m_p c^3 \Gamma} \right)^{1/3} r_0^{2/3} \\ &\simeq 4.9 \times 10^{11} L_{K,55}^{1/3} \left(\frac{\sigma_{e\gamma}}{\sigma_T} \right)^{1/3} \Gamma_{2.7}^{-1/3} r_{0,8}^{2/3} \text{ cm} \end{aligned} \quad (\text{B1})$$

for the unsaturated acceleration regime; for the saturated acceleration regime,

$$\begin{aligned} R_{ph} &\simeq \frac{L_K \sigma_{e\gamma}}{8\pi m_p c^3 \Gamma^3} \\ &\simeq 4.7 \times 10^{13} L_{K,55} \left(\frac{\sigma_{e\gamma}}{\sigma_T} \right) \Gamma_{2.7}^{-3} \text{ cm}. \end{aligned} \quad (\text{B2})$$

The observed time delay $\simeq (1+z)R_{ph}/2c\Gamma^2$ for $R_{ph} \lesssim 10^{11-13}$ cm relative to the keV to GeV component is less than a few milliseconds. This means that two components arrive almost simultaneously, which is consistent with the observation.

2) The synchrotron self-Compton (SSC) process in the internal shocks could also account for the two components in the prompt phase. The peaking MeV component comes from synchrotron emission of the accelerated electrons, while some of them behave as seed photons for the IC scattering. However, peak MeV emission should not be quasi-thermal since $\tau_{KN} < 1$, which is not consistent with the modeling results that the mBB function is better than the BAND function in describing peak MeV emission.

Thus, we think that the first explanation is more consistent with the observations.

Appendix C: The estimates of Γ

1. Estimate of the Range of Γ with quasi-thermal emission

The approach for the estimate of Γ is sketched in broad terms in the typical fireball model of GRBs [e.g. 56, 57] as follows: the thermal component originates from the photosphere of an expanding plasma jet; the observed thermal flux (integrated over all frequencies) is given by integrating the intensity over the emitting [58],

$$F_{BB}^{\text{obs}} = \frac{2\pi}{d_L^2} \int_{\mu_{\min}}^{\mu_{\max}} d\mu \mu r_{ph}^2 \mathcal{D}^4 (\sigma_{SB} T_{ph}'^4 / \pi). \quad (\text{C1})$$

Here, T_{ph}' is the comoving temperature at the photospheric radius, σ_{SB} is Stefan's constant and $\mathcal{D} = \mathcal{D}(\theta)$ is the Doppler factor, $\mathcal{D} = [\Gamma(1 - \beta\mu)]^{-1}$. The angle θ is the angle between the direction of the outflow velocity vector (β) and the line of sight, $\mu \equiv \cos(\theta)$, and $\Gamma = (1 - \beta^2)^{-1/2}$ is the outflow Lorentz factor. The observed radiation is dominated by photons emitted close to the line of sight ($\theta_{\min} = 0$), the upper integration boundary in the equation for the thermal flux is $\mu_{\max} = \cos \theta_{\min} = 1$, thus one has [58, 59]:

$$T_{\text{obs}} \simeq C_1 \Gamma_{ph} T_{ph}' / (1 + z), \quad (\text{C2})$$

$$F_{BB}^{\text{obs}} \simeq \mathcal{R}^2 \sigma_{SB} T_{\text{obs}}^4, \quad (\text{C3})$$

where

$$\mathcal{R} = C_2 \frac{(1+z)^2}{d_L} \frac{r_{ph}}{\Gamma_{ph}}; \quad (\text{C4})$$

$C_1 \simeq 1.48$ and $C_2 \simeq 1.06$ are factors derived from the detailed numerical integration of the angle- and distance-dependent photosphere emission.

Taking into account different regimes of acceleration of the outflow, for example, in the case of saturation regime (i.e. r_{ph} is greater than r_s , and r_s is the saturation radius), $r_{ph} = L\sigma_T/8\pi\Gamma^3 m_p c^3$ (here $L = 4\pi d_L^2 Y F^{\text{obs}}$, $Y \geq 1$ is the ratio between the total energy and the energy emitted in γ -rays, F^{obs} is the sum of thermal and non-thermal component; $\Gamma = \eta$, where η is the dimensionless entropy), one has

$$\Gamma = \left[(1.06) (1+z)^2 d_L \frac{Y F^{\text{obs}} \sigma_T}{2m_p c^3 \mathcal{R}} \right]^{1/4}. \quad (\text{C5})$$

More realistically, the observed spectra are broader than the Planck spectrum, because of geometrical broadening [e.g. 28]

This means that the observed spectrum is a superposition of a series of blackbodies of different temperatures arising from different angles to the line of sight; therefore, a multi-blackbody (mBB) function could be used to describe the quasi-thermal spectrum, and the temperatures of the

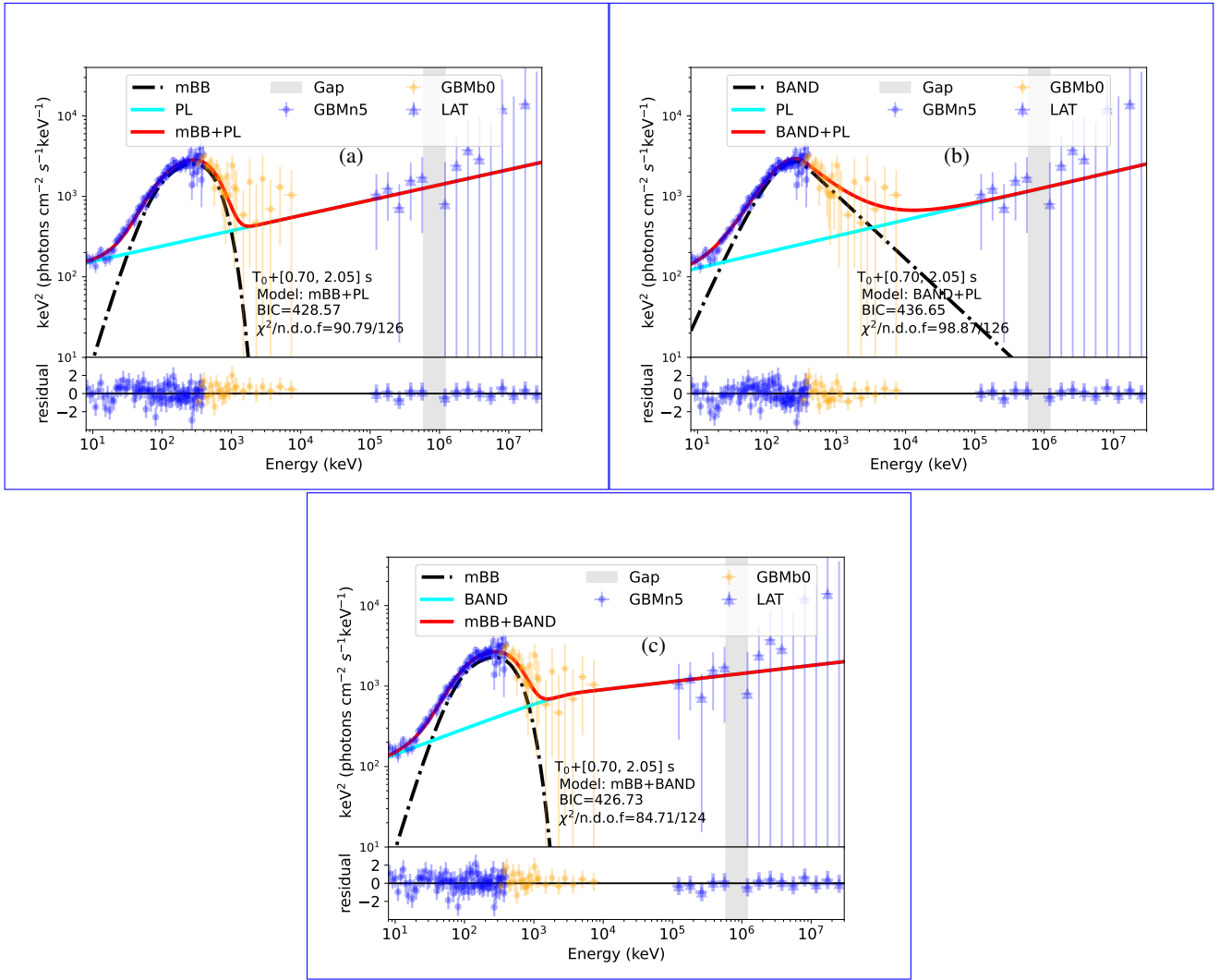


FIG. 3. (Color online) Modeling results with mBB+BAND, BAND, mBB+BAND, PL and mBB+BAND + PL, with considering zero-count in bins inside the gap (marked in gray) are excluded from the fits.

individual Planck function are related by a power law. In addition, the other component of the outflow (i.e., Poynting flux) also affects the estimate of Γ . Therefore, this approach is extended based on basic physical models (the so-called ‘top-down’ approach; see the details in [59] and [33]). Note that in an unsaturated regime, Γ , r_{ph} are related to the initial radius r_0 , physically $r_0 = 10^7 - 10^9$ cm, therefore the range of Γ value is estimated.

2. Estimation based on statistical correlation

There are empirical correlations between Γ and E_{iso} and L_{iso} :

3. The estimate of Γ in GRB 210704A

The ranges of Γ is estimated in different regimes using the ‘top-down’ approach. Regime II ($r_{\text{ra}} < r_{\text{ph}} < r_{\text{c}}$, where r_{ra} is the rapid acceleration radius, and r_{c} is the coasting radius) is allowed. For $T_0 + [0.70, 2.05]$ s and assuming $Y = 10$, Γ ranges from 500 to 900. Note that from Eq (C5), $\Gamma \propto Y^{1/4}$, the value of Y does not affect the estimate of Γ much, e.g., the factor of $Y^{1/4}$ of $Y = 10$ is 1.5 times that of $Y = 2$.

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